A FAST ITERATIVE SHRINKAGE-THRESHOLDING ALGORITHM FOR LINEAR INVERSE PROBLEMS

22MAT202

Group - 12
Lakshmi Tejaswi - CB.SC.U4AIE24116
K.Kiranmai -CB.SC.U4AIE24119
K. Sudeshna-CB.SC.U4AIE24120

INTRODUTION

Images can become blurred due to factors like camera motion, defocusing, or object movement, which reduces clarity and fine details. This project focuses on recovering sharp images from blurred ones using the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA). By leveraging gradient updates and thresholding, it produces high-quality deblurred images, improving their usability for applications like photography.

PROBLEM STATEMENT

Images frequently become blurred during capture or transmission due to factors like camera shake, defocusing, or motion. This blur reduces image clarity and can obscure important details, making analysis difficult in applications such as photography. Recovering a sharp and clear image from a blurred one is essential to improve visual quality and enable accurate interpretation. The challenge lies in removing the blur while preserving important structures and details in the image.

LITERATURE REVIEW-1

A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems

Methodology:

- Solve linear inverse problems
- Use proximal operators for nonsmooth terms.
- Apply backtracking line search to adapt step size
- Applications: image deblurring, compressed sensing, sparse signal recovery.

Limitations:

- Only guaranteed for convex problems.
- Sensitive to high noise in measurements.
- Can be computationally heavy for large images

LITERATURE REVIEW-2

Structured FISTA for Image Restoration

Methodology:

- The paper extends FISTA by using structured operators like blur operators and wavelets to speed up image restoration while keeping accuracy.
- It applies iterative shrinkage and thresholding in the wavelet domain, reducing noise and improving sharpness in restored images

Limitation:

Structured FISTA depends on choice of blur operators and wavelets → performance drops if these are not well-matched to the image.

LITERATURE REVIEW-3

A New TwIST: Two-Step Iterative Shrinkage/ Thresholding Algorithms for Image Restoration(2007)

Methodology:

- Extends standard IST (Iterative Shrinkage-Thresholding) by using a two-step update rule instead of a single step → accelerates convergence.
- Combines previous two iterates to compute the new one

Limitations:

- Works best only if image is sparse in transform domain (wavelets, etc.) → not universally effective.
- dependency on sparsity model

METHODOLOGY

Input Image Preprocessing

- Load the input image and convert to floating-point representation.
- Handle color channels (RGB) separately.

Blur Simulation

- Define a Gaussian-like kernel.
- Apply convolution (convolve2d) with symmetric boundary conditions.

Optimization Setup

- Formulate the deblurring problem as a sparse regularization task.
- Define forward operator (blur: A) and adjoint operator (At).

FISTA with Backtracking

- Iteratively recover sharp image using soft-thresholding and adaptive line search.
- Exploit momentum update for faster convergence.

Evaluation & Visualization

- Compute PSNR (Peak Signal-to-Noise Ratio) before and after restoration.
- Display original, blurred, and reconstructed images side by side.

ALGORITHM EXPLANTION

Fast Iterative Shrinkage-Thresholding Algorithm (FISTA)

- Problem Formulation
  - We solve:

$$\min_x \frac{1}{2}\|Ax - b\|^2 + \lambda\|x\|_1$$

    where
    - A: blur operator
    - b: blurred image
    - λ: regularization (sparsity control)
- Key Steps
  1. Gradient Descent Step
     - Compute gradient:

$$\nabla f(y) = A^T(Ay - b)$$

  2. Proximal Update (Soft-Thresholding)
     - Enforces sparsity in the solution.
- Backtracking Line Search
  - Adaptively chooses the step size L for stable and faster convergence.

ALGORITHM EXPLANATION

Working of FISTA with Acceleration

- Momentum Update
  - Introduce auxiliary variable yk to accelerate convergence.
  - Update rule:

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}$$

$$y_{k+1} = x_{k+1} + \frac{t_k - 1}{t_{k+1}}(x_{k+1} - x_k)$$

- Advantages of FISTA
  - Faster convergence than ISTA (Order O(1/k^2)).
  - Produces sharper reconstructed images with fewer iterations.
  - Automatically balances deblurring and noise removal via λ.
- Output
  - Reconstructed image with higher PSNR compared to blurred image.

FUTURE WORK

While our current work focuses on classical FISTA-based image deblurring, future research can explore the integration of Plug-and-Play (PnP) deep denoisers. By replacing the traditional soft-thresholding step with advanced learned denoisers such as DnCNN or diffusion-based priors, the method can achieve improved perceptual quality while still preserving the underlying structure of FISTA.

THANK YOU